

Yang Hu

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EDUCATION

Tsinghua University, Beijing, China

08/2022–06/2026

B.Eng. in Computer Science, Yao Class, Institute for Interdisciplinary Information Sciences

RESEARCH VISITS

• **Carnegie Mellon University**

Spring 2025

Student intern advised by William Kuszmaul

PUBLICATIONS

Authors are in alphabetical order.

Dynamic Entropy-Encoded Arrays in $O(1)$ Time with Nearly Optimal Space

Guy E. Blelloch, **Yang Hu**, William Kuszmaul, Tianxiao Li, Renfei Zhou.

In *Proc. FOCS*, 2026 (to appear).

[k-Clustering via Iterative Randomized Rounding](#)

Jarosław Byrka, Yuhao Guo, **Yang Hu**, Shi Li, Chengzhang Wan, Zaixuan Wang.

In *Proc. FOCS*, 2026 (to appear).

[Quadratic Probing Revisited: Smoothed Analysis and the Fall of Robin Hood](#)

Yang Hu, William Kuszmaul, Jingxun Liang, Stefan Walzer, Huacheng Yu, Renfei Zhou.

[Fast Concurrent Primitives Despite Contention](#)

Michael A. Bender, Guy E. Blelloch, Martín Farach-Colton, **Yang Hu**, Rob Johnson, Rotem Oshman, Renfei Zhou.

In *Proc. SPAA*, 2026.

[New Algebrization Barriers to Circuit Lower Bounds via Communication Complexity of Missing-String](#)

Lijie Chen, **Yang Hu**, Hanlin Ren.

In *Proc. ITCS*, 2026.

[Nearly Optimal Bounds for Stochastic Online Sorting](#)

Yang Hu.

In *Proc. SODA*, 2026.

[Static Retrieval Revisited: To Optimality and Beyond](#)

Yang Hu, William Kuszmaul, Jingxun Liang, Huacheng Yu, Junkai Zhang, Renfei Zhou.

In *Proc. FOCS*, 2025.

[Nash Social Welfare with Submodular Valuations: Approximation Algorithms and Integrality Gaps](#)

Xiaohui Bei, Yuda Feng, **Yang Hu**, Shi Li, Ruilong Zhang.

In *Proc. STOC*, 2026.

[Constant Approximation for Weighted Nash Social Welfare with Submodular Valuations](#)

Yuda Feng, **Yang Hu**, Shi Li, Ruilong Zhang.

In *Proc. STOC*, 2025.

[Optimal Static Dictionary with Worst-Case Constant Query Time](#)

Yang Hu, Jingxun Liang, Huacheng Yu, Junkai Zhang, Renfei Zhou.

In *Proc. STOC*, 2025.

SELECTED AWARDS AND SCHOLARSHIPS

- **Yao Award (Silver Medal)** 2025
Awarded to 3 students in Yao Class each year

COMPETITIVE PROGRAMMING (SELECTED)

- **The 2022 ICPC Asia East Continent Final Contest** 2023
Gold award, 4th place (with teammates Junkai Zhang and Hao Hu)
- **ACM International Collegiate Programming Contest, Asia Regional Contest, Nanjing Site** 2022
Gold medal, 3rd place (solo)
- **ACM International Collegiate Programming Contest, Asia Regional Contest, Jinan Site** 2022
Gold medal, 2nd place (with teammates Junkai Zhang and Hao Hu)
- **China National Olympiad in Informatics** 2021
Gold medal, 3rd place
- **The Asia-Pacific Informatics Olympiad** 2021
Ranked 4th in China
- **CCF NOI Winter Camp** 2021
Gold medal, 1st place

SERVICE

- **Algorithm Analysis and Design (Tsinghua University)** Autumn 2024
Teaching assistant (Instructor: Ran Duan)

SKILLS

- TOEFL iBT Score 115 02/2024
- Japanese N1 06/2022